



Alchemists of the Data Era

A new breed of data superhero has arrived, and their numbers are growing by the day. We tell you all about them and how to become one.

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Imagine you are travelling to work. Your car's navigation unit tells you what route to take based on where everyone else is going. You need to pay your insurance deposit this month. Little do you know that your insurance provider has already predicted the likelihood of a claim based on user and vehicle characteristics. You catch up with email and social media updates on the way. The ads you see have been designed to get the highest clicks based on preferences of million of people. You read though travel reviews for a trip you are planning, and wonder how the most inspiring photos seem to always make it to the top of a review website. You think about work. Your company has to decide how much to bid on a new project, and you need data that will tell you what your competitors are likely to bid. You have a doctor's appointment first, but your doctor has already predicted the probability of you getting sick. Your local government health organisation can tell who is more at risk based on Google searches. You need to visit the grocery store after work. Luckily your supermarket has predicted what customers are going to visit on a particular day and have stocked their aisles accordingly.

This is the era of Big Data, where petabytes of data are being generated because of the scale of tools available to generate them, and because storage is cheaper than ever. Using all this data scientifically in complex ways to answer questions and improve services is what Data Science is all about. Data Science is essentially what moves the world around these days. It spans domains like the environmental and behavioural sciences, healthcare, insurance, genomics, marketing, finance, operations, and technology. Organisations rely on ever improving algorithms



New scientific data techniques are revolutionising how the world works

and computational models to predict trends and improve products and services. They use computational models to identify the components of a problem and the relationships between them. They use data analysis tools to understand patterns, boundaries, estimate hidden parameters, and predict and retrodict from data. These tools and techniques can also help examine the consequences of unexpected events, tell you when to intervene, which choice is best, what happens when you make a choice, and what happens if you do not.

What isn't Data Science?

Data Science has, too an extent, become an umbrella term for all activities that deal with data, but which are not necessarily scientific. This has led to a debate about what Data Science truly is. Not everyone who works with data is a Data Scientist.

Merely gathering or processing data does not count. So Database Administrators, Data Architects, Systems Engineers

and Application Developers, who deal exclusively with tools like Hadoop and noSQL, are not automatically Data Scientists, though Data Scientists could make use of these tools as well. After all, you need to gather and clean data before you analyse it. Business Intelligence (BI) professionals who use tools like SAP are not Data Scientists either, as they merely create dashboards and reports based on pre-conceived notions of how their data is supposed to look. They do not investigate or ask any new questions of their data that require new modelling or analytical techniques.

A true Data Scientist, according to experts, is a cross between software engineer and statistician. It is someone who investigates hypotheses about a data set, and then uses advanced algorithms and applied statistical applications to verify these hypotheses and make inferences. To use these algorithms, they need knowledge of math and stats, combined with computer science and domain knowledge.